

SOFTWARE ENGINEERING

Academic Elective Bidding & Allocation System

LAB 1

NAME	Pratham S Reddy
SRN	PES1UG24AM200
DATE	20/8/26
PROBLEM STATEMENT	#05 - Academic Elective Bidding & Allocation System

Deliverable 1: Requirements Table

Functional Requirements:

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow students to distribute 100 bidding credits across their ranked elective preferences and validate prerequisite course completions.	High	Pass: Credits deduct accurately and prerequisites are validated. Fail: Total allocated credits exceed 100 or unmet prerequisites are bypassed.	Ensures valid bidding and prerequisite compliance.
FR-002	Functional	The system shall allow students to view available elective courses with course details and eligibility requirements before submitting bids.	High	Pass: Eligible electives and requirements are displayed correctly. Fail: Required course information is missing or incorrect.	Helps students make informed elective choices.
FR-003	Functional	The system shall validate submitted bids and prevent bids that exceed available credits or violate prerequisite requirements.	High	Pass: Invalid bids are rejected with clear feedback. Fail: An invalid bid is accepted for allocation.	Prevents invalid data from entering the allocation process.
FR-004	Functional	The system shall allocate electives based on student bids while resolving allocation constraints and preventing	High	Pass: Allocation satisfies bidding and schedule constraints. Fail: A student receives conflicting electives or	Ensures the final allocation is feasible and conflict free.

		timetable collisions.		constraints are violated.	
FR-005	Functional	The system shall allow students to view their final elective allocation results after the allocation process is completed.	Medium	Pass: Students can view their allocated electives accurately. Fail: Results are unavailable or incorrect.	Provides students with their final elective selection.

Non Functional Requirements:

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Nonfunctional: Performance and Security	The final elective allocation solver shall process 5,000 student bids and resolve schedule conflicts in under 30 seconds while maintaining required security standards.	High	Pass: Benchmarking confirms processing within 30 seconds and security standards are maintained under simulated peak load. Fail: Processing exceeds 30 seconds or security requirements are not met.	Ensures responsive and reliable allocation during peak bidding periods.
NFR-002	Nonfunctional: Reliability and Data Integrity	The system shall maintain accurate student bidding and allocation data throughout the process and prevent loss or corruption of submitted bids.	High	Pass: Submitted bids and final allocations remain accurate after processing. Fail: Bids are lost, duplicated, corrupted, or associated with the wrong student.	Protects the accuracy and integrity of student allocation data.

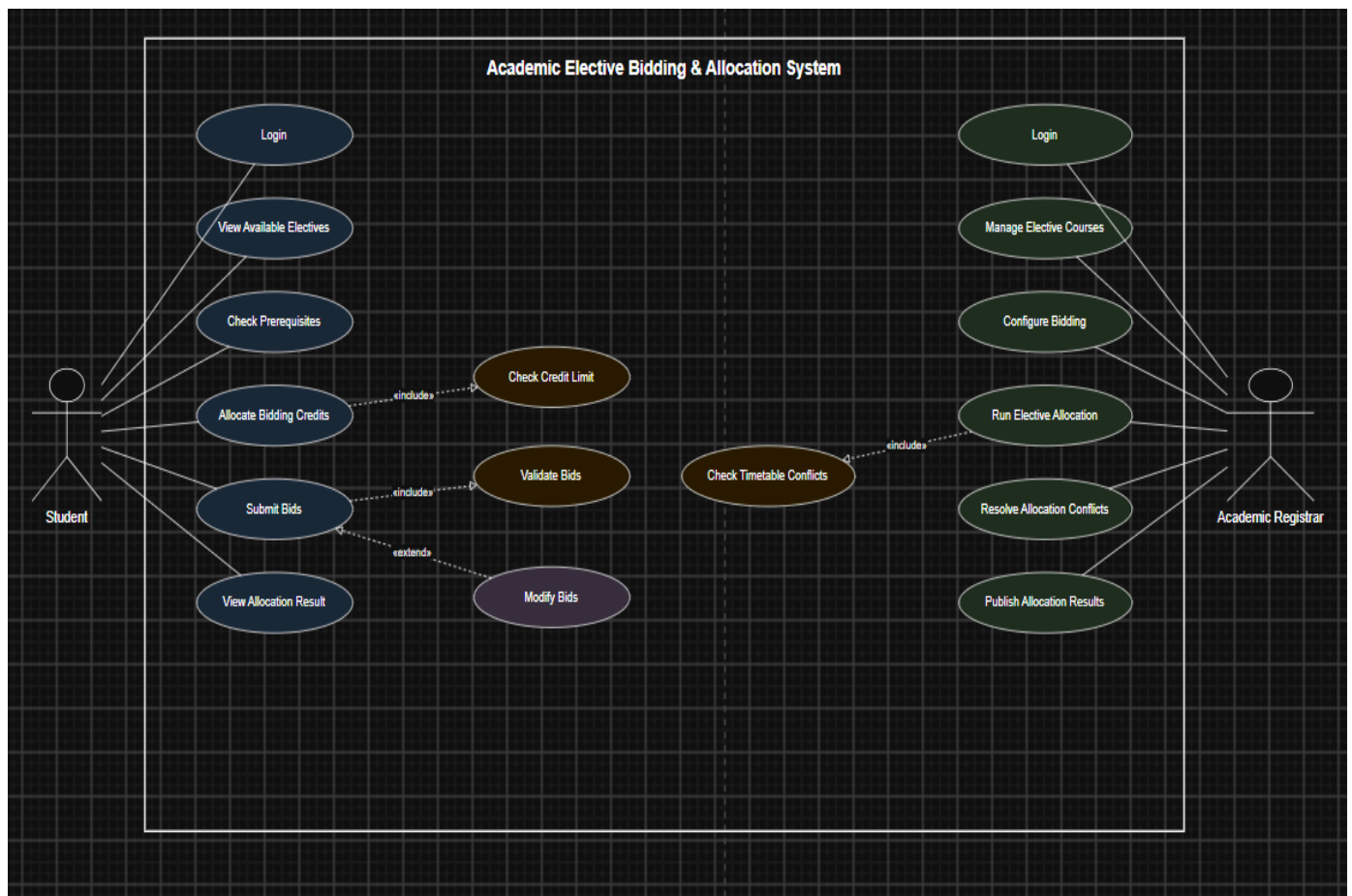
Deliverable 2: UML Use Case Diagram

The UML diagram identifies the Student and Academic Registrar actors, their primary use cases, supporting use cases, and the required include and extend relationships.

Actor	Primary Use Cases	Secondary Use Cases
Student	Login; View Available Electives; Allocate Bidding Credits; Submit Bids; View Allocation Result	Check Prerequisites; Validate Bids; Check Credit Limit; Modify Bids
Academic Registrar	Login; Manage Elective Courses; Configure Bidding; Run Elective Allocation; Publish Allocation Results	Resolve Allocation Conflicts; Check Timetable Conflicts

UML Relationships

Relationship	Source	Target
include	Submit Bids	Validate Bids
include	Allocate Bidding Credits	Check Credit Limit
include	Run Elective Allocation	Check Timetable Conflicts
extend	Modify Bids	Submit Bids



Deliverable 3: Use Case Flow Specification

Field	Value
Use Case ID	UC-01
Use Case Name	Submit Elective Bids
Primary Actor	Student
Secondary Actors	Academic Registrar, System
Description	The student selects elective preferences, distributes bidding credits, and submits the bid. The system validates credits and prerequisites before storing the bid.
Trigger	Student clicks Submit Bids after selecting and ranking elective preferences.

Preconditions

Student is authenticated.

The elective bidding period is open.

Student has completed the required prerequisites for selected electives.

Student has bidding credits available.

Postconditions

Valid bid is stored successfully and the allocated bidding credits are updated.

Submitted preferences are available to the allocation process.

If validation fails, the invalid bid is not submitted and the student receives an error message.

Main Success Scenario

Step	Actor	Action
1	Student	Selects and ranks elective preferences.
2	System	Displays available bidding credit balance.
3	Student	Distributes bidding credits across the ranked electives.
4	System	Validates that total allocated credits do not exceed 100.
5	System	Checks prerequisite completion for selected electives.
6	Student	Confirms and submits the bid.
7	System	Stores the bid and updates the student's credit allocation.

8	System	Displays a successful submission confirmation.
9	Use Case	Ends successfully.

Alternate Flow A1: Credit Limit Exceeded

4a. System detects that total allocated bidding credits exceed 100.

4a.1. System displays the credit limit error.

4a.2. Student adjusts the credit distribution.

4a.3. System resumes at Step 4 and validates the corrected bid.

Alternate Flow A2: Prerequisite Not Satisfied

5a. System detects that a selected elective has an unmet prerequisite.

5a.1. System rejects the affected preference and displays the prerequisite requirement.

5a.2. Student removes the ineligible elective or selects another eligible elective.

5a.3. System resumes at Step 5 and revalidates prerequisites.